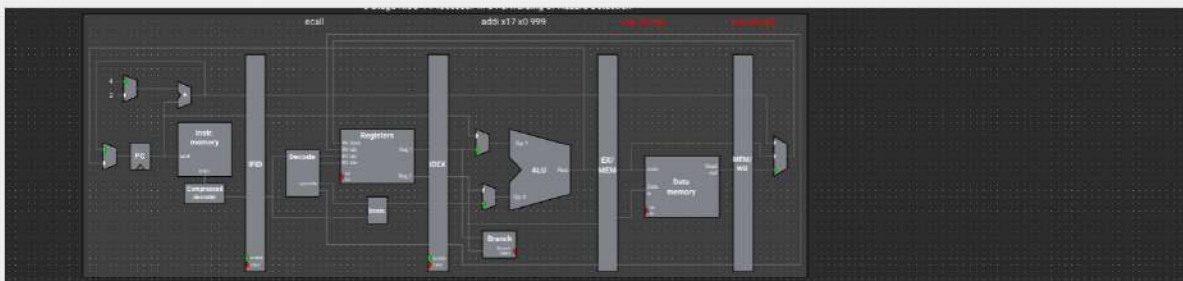


File Edit View Help

100 ms



Console Memory

Console

Execution info

Cycles: 10
Instrs. retired: 6
CPI: 1.67
IPC: 0.6
Clock rate: 0 Hz

Registers

gpr		
Name	Alias	Value
x0	zero	0x00000000
x1	ra	0x00000000
x2	sp	0x7fffffff
x3	gp	0x10000000
x4	tp	0x00000000
x5	t0	0x00000000
x6	t1	0x00000000
x7	t2	0x00000000
x8	s0	0x00000000

Display type: Hex

Instruction memory

BP	Addr	Stage	Instruction
☐	0x0		auipc x5 ...
☐	0x4		addi x5 x5 0
☐	0x8		lw x17 0 x5
☐	0xc		add x17 x1...
☐	0x10		add x17 x1...
☐	0x14		hwp x17 x1

Processor: 5-stage processor w/o forwarding or hazard detection ISA: RV32IM

File Edit View Help

100 ms

100
1010
9+
Editor

Processor

Cache

Memory

I/O

Source code

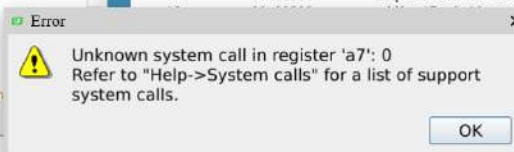
Input type: ☒ Assembly ☐ Executable code

View mode: ☐ Binary ☒ Disassembled

gpr

```
1 .data
2 val: .word 10
3
4 .text
5 .globl main
6
7 main:
8     la x5, val
9     lw x17, 0(x5)
10
11     add x17, x17, x0
12     add x17, x17, x0
13
14     beq x17, x17, wrong # ALWAYS TRUE -> CONTROL H
```

```
00000000 <main>:
0:      10000297      auipc x5 0x10000
4:      00026293      addi x5 x5 0
8:      0002a883      lw x17 0 x5
c:      000888b3      add x17 x17 x0
10:     000888b3      add x17 x17 x0
14:     01188663      beq x17 x17 12 <wrong>
```



Name	Alias	Value
x0	zero	0x00000000
x1	ra	0x00000000
x2	sp	0x7fffffff0
x3	gp	0x10000000
x4	tp	0x00000000
x5	t0	0x00000000
x6	t1	0x00000000
x7	t2	0x00000000
x8	s0	0x00000000
x9	s1	0x00000000
x10	a0	0x00000000
x11	a1	0x00000000
x12	a2	0x00000000
x13	a3	0x00000000
x14	a4	0x00000000
x15	a5	0x00000000
x16	a6	0x00000000

Display type: Hex

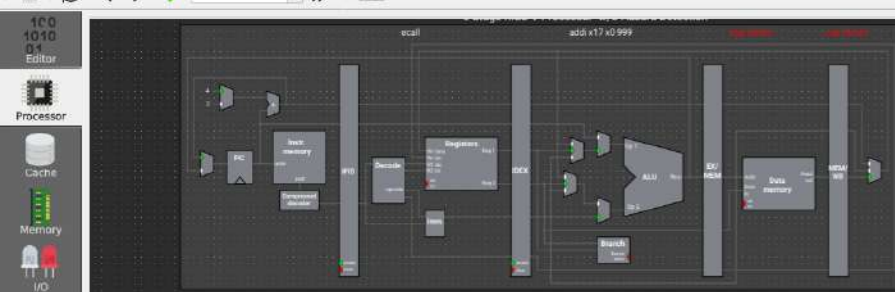
Processor: 5-stage processor w/o forwarding or hazard detection ISA: RV32IM

Qt Pipeline diagram



of recorded cycles can be changed in settings

	0	1
auipc x5 0x10000	IF	ID
addi x5 x5 0		IF
lw x17 0 x5		
add x17 x17 x0		
add x17 x17 x0		
beq x17 x17 12 <wrong>		
addi x17 x0 10		
ecall		
addi x17 x0 999		
ecall		



Name	Alias	Value
x0	zero	0x00000000
x1	ra	0x00000000
x2	sp	0x7fffffff0
x3	gp	0x10000000
x4	tp	0x00000000
x5	t0	0x10000000
x6	t1	0x00000000
x7	t2	0x00000000
x8	s0	0x00000000

Instruction memory

BP	Addr	Stage	Instruction
☐	0x0		auipc x5 ...
☐	0x4		addi x5 x5 0
☐	0x8		lw x17 0 x5
☐	0xc		add x17 x1...
☐	0x10		add x17 x1...
☐	0x14		beq x17 x1...

Cycles:	10
Instrs. retired:	6
CPI:	1.67
IPC:	0.6
Clock rate:	0 Hz

Console

Console

File Edit View Help

100 ms

Source code

```
1 .data
2 val: .word 10
3
4 .text
5 .globl main
6
7 main:
8     la x5, val
9     lw x17, 0(x5)
10
11     add x17, x17, x0
12     add x17, x17, x0
13
14     beq x17, x17, wrong # ALWAYS TRUE -> CONTROL h
```

Input type: ☒ Assembly ☐ Executable code View mode: ☐ Binary ☒ Disassembled

gpr

Name	Alias	Value
x0	zero	0x00000000
x1	ra	0x00000000
x2	sp	0x7fffffff0
x3	gp	0x10000000
x4	tp	0x00000000
x5	t0	0x10000000
x6	t1	0x00000000
x7	t2	0x00000000
x8	s0	0x00000000
x9	s1	0x00000000
x10	a0	0x00000000
x11	a1	0x00000000
x12	a2	0x00000000
x13	a3	0x00000000
x14	a4	0x00000000
x15	a5	0x00000000
x16	a6	0x00000000

Display type: Hex

Processor: 5-stage processor w/o hazard detection ISA: RV32IM

Error

Unknown system call in register 'a7': 268435456
Refer to "Help->System calls" for a list of support system calls.

OK

Qt

Pipeline diagram

X



of recorded cycles can be changed in settings

	0	1
auipc x5 0x10000	IF	ID
addi x5 x5 0		IF
lw x17 0 x5		
add x17 x17 x0		
add x17 x17 x0		
beq x17 x17 12 <wrong>		
addi x17 x0 10		
ecall		
addi x17 x0 999		
ecall		

File Edit View Help

100 ms



Console Memory

Console

Execution info

Cycles: 23
Instrs. retired: 7
CPI: 3.29
IPC: 0.304
Clock rate: 0 Hz

Registers

gpr		
Name	Alias	Value
x0	zero	0x00000000
x1	ra	0x00000000
x2	sp	0x7fffffff0
x3	gp	0x10000000
x4	tp	0x00000000
x5	t0	0x10000000
x6	t1	0x00000000
x7	t2	0x00000000
x8	s0	0x00000000

Display type: Hex

Instruction memory

BP	Addr	Stage	Instruction
☐	0x0		auipc x5 ...
☐	0x4		addi x5 x5 0
☐	0x8		lw x17 0 x5
☐	0xc		add x17 x1...
☐	0x10		add x17 x1...
☐	0x14		beq x17 x1...

Processor: 5-Stage processor w/o forwarding unit ISA: RV32IM

File Edit View Help

100 ms

100
1010
0*
Editor

Processor

Cache

Memory

I/O

Source code

Input type: ☒ Assembly ☐ Executable code

View mode: ☐ Binary ☒ Disassembled

gpr

```
1 .data
2 val: .word 10
3
4 .text
5 .globl main
6
7 main:
8     la x5, val
9     lw x17, 0(x5)
10
11     add x17, x17, x0
12     add x17, x17, x0
13
14     beq x17, x17, wrong # ALWAYS TRUE -> CONTROL h
```

```
00000000 <main>:
0:      10000297      auipc x5, 0x10000
4:      00028293      addi x5, x5, 0
8:      0002a883      lw x17, 0(x5)
c:      000888b3      add x17, x17, x0
10:     000888b3      add x17, x17, x0
14:     01188663      beq x17, x17, 12 <wrong>
```

Error

Unknown system call in register 'a7': 999
Refer to "Help->System calls" for a list of support
system calls.

OK

Name	Alias	Value
x0	zero	0x00000000
x1	ra	0x00000000
x2	sp	0x7fffffff0
x3	gp	0x10000000
x4	tp	0x00000000
x5	t0	0x10000000
x6	t1	0x00000000
x7	t2	0x00000000
x8	s0	0x00000000
x9	s1	0x00000000
x10	a0	0x00000000
x11	a1	0x00000000
x12	a2	0x00000000
x13	a3	0x00000000
x14	a4	0x00000000
x15	a5	0x00000000
x16	a6	0x00000000

Display type: Hex

Processor: 5-Stage processor w/o forwarding unit ISA: RV32IM

Pipeline diagram



of recorded cycles can be changed in settings

	0	1
auipc x5 0x10000	IF	ID
addi x5 x5 0		IF
lw x17 0 x5		
add x17 x17 x0		
add x17 x17 x0		
beq x17 x17 12 <wrong>		
addi x17 x0 10		
ecall		
addi x17 x0 999		
ecall		